

Assignment 7: GLMs (Linear Regressios, ANOVA, & t-tests)

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A07_GLMs.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

!! AI generated code should NOT be used in this assignment

Set up your session

1. Set up your session. Check your working directory. Load the tidyverse, agricolae and other needed packages. Import the *raw* NTL-LTER raw data file for chemistry/physics (NTL-LTER_Lake_ChemistryPhysics_Raw.csv). Set date columns to date objects.
2. Build a ggplot theme and set it as your default theme.

```
#1
# Load necessary packages
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.3      v tidyr     1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```

library(agricolae)
library(here)

## here() starts at /home/guest/EDE_Fall2025

# check working directory
here()

## [1] "/home/guest/EDE_Fall2025"

# import raw NTL-LTER data file
ntl_lter_chem_physics <- read.csv(here("Data/Raw/NTL-LTER_Lake_ChemistryPhysics_Raw.csv"))

# set sample date to date object
ntl_lter_chem_physics$sampldate <- mdy(ntl_lter_chem_physics$sampldate)

#2
# Build ggplot theme
mytheme <- theme_classic() +
  theme(
    # set legend position
    legend.position = "bottom",

    # set fonts
    plot.title = element_text(size = 14),
    axis.title = element_text(size = 12),
    axis.text = element_text(size = 10, color = "black"),
    legend.text = element_text(size = 10, color = "black")
  )

# set custom theme as default
theme_set(mytheme)

```

Simple regression

Our first research question is: Does mean lake temperature recorded during July change with depth across all lakes?

3. State the null and alternative hypotheses for this question:

Answer: H0: mean lake temperature recorded in July does not change with depth across all lakes
 Ha: mean lake temperature recorded in July does change with depth across all lakes

4. Wrangle your NTL-LTER dataset with a pipe function so that the records meet the following criteria:
 - Only dates in July.
 - Only the columns: `lakename`, `year4`, `daynum`, `depth`, `temperature_C`
 - Only complete cases (i.e., remove NAs)

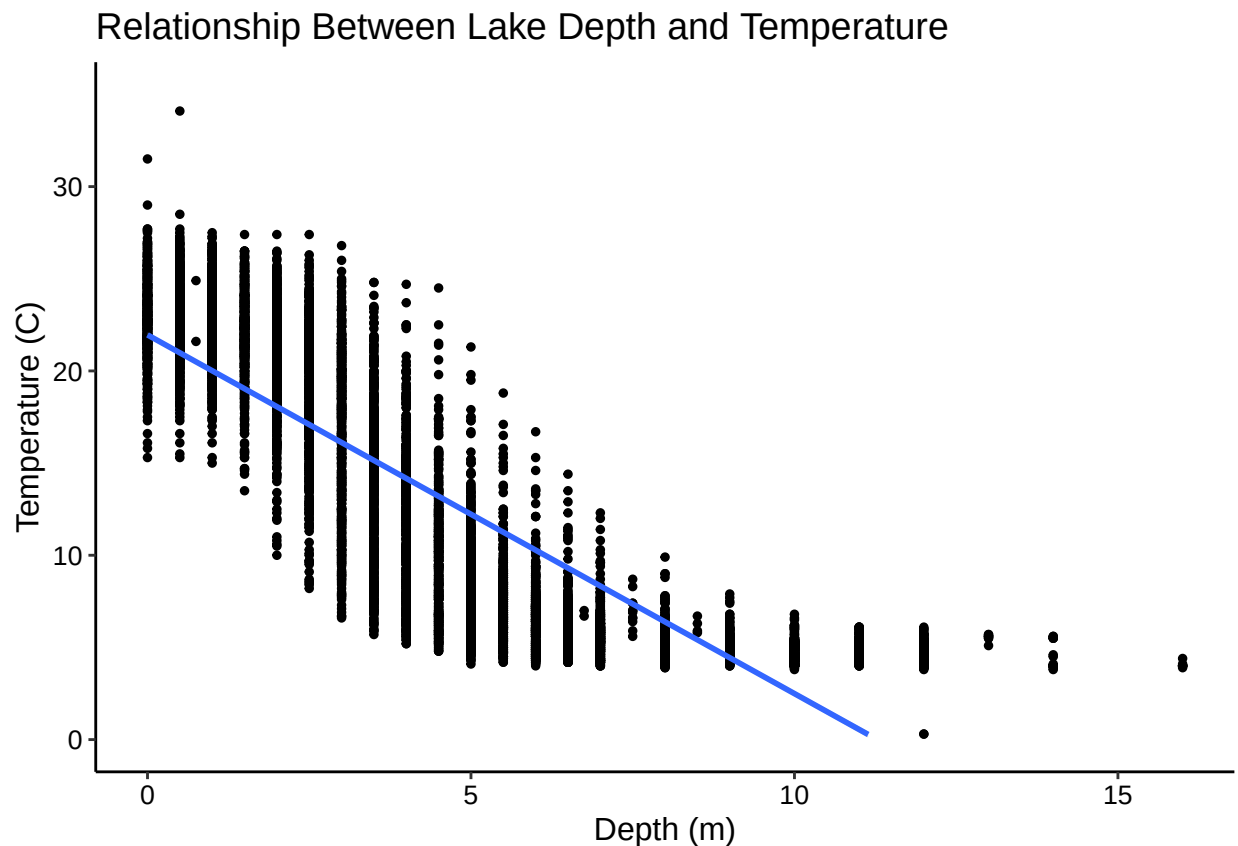
5. Visualize the relationship among the two continuous variables with a scatter plot of temperature by depth. Add a smoothed line showing the linear model, and limit temperature values from 0 to 35 °C. Make this plot look pretty and easy to read.

```
#4
ntl_lter_subset <- ntl_lter_chem_physics %>%
  filter(month(sampledate) == 7) %>%
  select(lakename, year4, daynum, depth, temperature_C) %>%
  na.omit()

#5
ggplot(ntl_lter_subset, aes(x = depth, y = temperature_C)) +
  geom_point(size = 1) +
  geom_smooth(method = lm) +
  scale_y_continuous(limits = c(0, 35)) +
  labs(x = "Depth (m)",
       y = "Temperature (C)",
       title = "Relationship Between Lake Depth and Temperature")
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 24 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```



6. Interpret the figure. What does it suggest with regards to the response of temperature to depth? Do the distribution of points suggest about anything about the linearity of this trend?

Answer: The figure suggests that there is a negative relationship between temperature and depth. At a certain depth, the temperature levels out around 5 degrees C, which might suggest a non-linear relationship. It could be helpful to plot depth vs. temperature on a logarithmic scale to see if a non-linear relationship fits better.

7. Perform a linear regression to test the relationship and display the results.

```
#7
# regression
temp_depth_regression <- lm(data = ntl_lter_subset,
                             temperature_C ~ depth)
summary(temp_depth_regression)

##
## Call:
## lm(formula = temperature_C ~ depth, data = ntl_lter_subset)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.5173 -3.0192  0.0633  2.9365 13.5834
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  21.95597    0.06792   323.3  <2e-16 ***
## depth       -1.94621    0.01174  -165.8  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.835 on 9726 degrees of freedom
## Multiple R-squared:  0.7387, Adjusted R-squared:  0.7387
## F-statistic: 2.75e+04 on 1 and 9726 DF, p-value: < 2.2e-16
```

8. Interpret your model results in words. Include how much of the variability in temperature is explained by changes in depth, the degrees of freedom on which this finding is based, and the statistical significance of the result. Also mention how much temperature is predicted to change for every 1m change in depth.

Answer: The depth coefficient in the regression is -1.95, which confirms there is a negative correlation: as depth increases 1 meter, we expect temperature to decrease 1.95 degrees C. The intercept is 21.96, which means we expect temperature to be 21.96 degrees C at the surface. The regression has an R^2 of 0.739, which means changes in depth can explain 73.9% of the variability in temperature. These findings are based on 9726 degrees of freedom (9728 observations) and are statistically significant with a p-value $< 2.2e-16$.

Multiple regression

Let's tackle a similar question from a different approach. Here, we want to explore what might the best set of predictors for lake temperature in July across the monitoring period at the North Temperate Lakes LTER.

9. Run an AIC to determine what set of explanatory variables (year4, daynum, depth) is best suited to predict temperature.
10. Run a multiple regression on the recommended set of variables.

```
#9
temp_regression_all <- lm(temperature_C ~ year4 + daynum + depth,
                          data = ntl_lter_subset)
step(temp_regression_all, direction = "backward")

## Start: AIC=26065.53
## temperature_C ~ year4 + daynum + depth
##
##           Df Sum of Sq    RSS   AIC
## <none>                 141687 26066
## - year4      1         101 141788 26070
## - daynum     1         1237 142924 26148
## - depth      1        404475 546161 39189

##
## Call:
## lm(formula = temperature_C ~ year4 + daynum + depth, data = ntl_lter_subset)
##
## Coefficients:
## (Intercept)      year4      daynum      depth
##    -8.57556     0.01134     0.03978    -1.94644
```

```
#10

recommended_regression <- lm(temperature_C ~ year4 + daynum + depth,
                             data = ntl_lter_subset)
summary(recommended_regression)

##
## Call:
## lm(formula = temperature_C ~ year4 + daynum + depth, data = ntl_lter_subset)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.6536 -3.0000  0.0902  2.9658 13.6123
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -8.575564   8.630715  -0.994  0.32044
## year4        0.011345   0.004299   2.639  0.00833 **
## daynum       0.039780   0.004317   9.215 < 2e-16 ***
## depth       -1.946437   0.011683 -166.611 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.817 on 9724 degrees of freedom
## Multiple R-squared:  0.7412, Adjusted R-squared:  0.7411
## F-statistic: 9283 on 3 and 9724 DF, p-value: < 2.2e-16
```

11. What is the final set of explanatory variables that the AIC method suggests we use to predict temperature in our multiple regression? How much of the observed variance does this model explain? Is this an improvement over the model using only depth as the explanatory variable?

Answer: The final set of explanatory variables suggested by the AIC method is year, date, and depth. This model explains 74.1% of the observed variance in temperature. This is a slight improvement over the model just using depth, which explained 73.9% of variance.

Analysis of Variance

12. Now we want to see whether the different lakes have, on average, different temperatures in the month of July. Run an ANOVA test to complete this analysis. (No need to test assumptions of normality or similar variances.) Create two sets of models: one expressed as an ANOVA models and another expressed as a linear model (as done in our lessons).

```
#12
# expressed as an ANOVA model
lake_temp_anova <- aov(temperature_C ~ lakename,
                      data = ntl_lter_subset)
summary(lake_temp_anova)

##              Df Sum Sq Mean Sq F value Pr(>F)
## lakename      8  21642   2705.2     50 <2e-16 ***
## Residuals    9719 525813     54.1
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# expressed as a linear model
lake_temp_anova2 <- lm(temperature_C ~ lakename,
                      data = ntl_lter_subset)
summary(lake_temp_anova2)

##
## Call:
## lm(formula = temperature_C ~ lakename, data = ntl_lter_subset)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.769   -6.614   -2.679    7.684   23.832
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      17.6664     0.6501  27.174 < 2e-16 ***
## lakenameCrampton Lake    -2.3145     0.7699  -3.006 0.002653 **
## lakenameEast Long Lake   -7.3987     0.6918 -10.695 < 2e-16 ***
## lakenameHummingbird Lake -6.8931     0.9429  -7.311 2.87e-13 ***
## lakenamePaul Lake        -3.8522     0.6656  -5.788 7.36e-09 ***
## lakenamePeter Lake       -4.3501     0.6645  -6.547 6.17e-11 ***
## lakenameTuesday Lake    -6.5972     0.6769  -9.746 < 2e-16 ***
```

```
## lakenameWard Lake      -3.2078      0.9429   -3.402 0.000672 ***
## lakenameWest Long Lake -6.0878      0.6895   -8.829 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.355 on 9719 degrees of freedom
## Multiple R-squared:  0.03953,    Adjusted R-squared:  0.03874
## F-statistic:    50 on 8 and 9719 DF,  p-value: < 2.2e-16
```

13. Is there a significant difference in mean temperature among the lakes? Report your findings.

Answer: the p-value of the ANOVA is $< 2.2e-16$, which means we can reject our null hypothesis (that there is no difference) and conclude that there is a significant difference in mean temperature among the lakes.

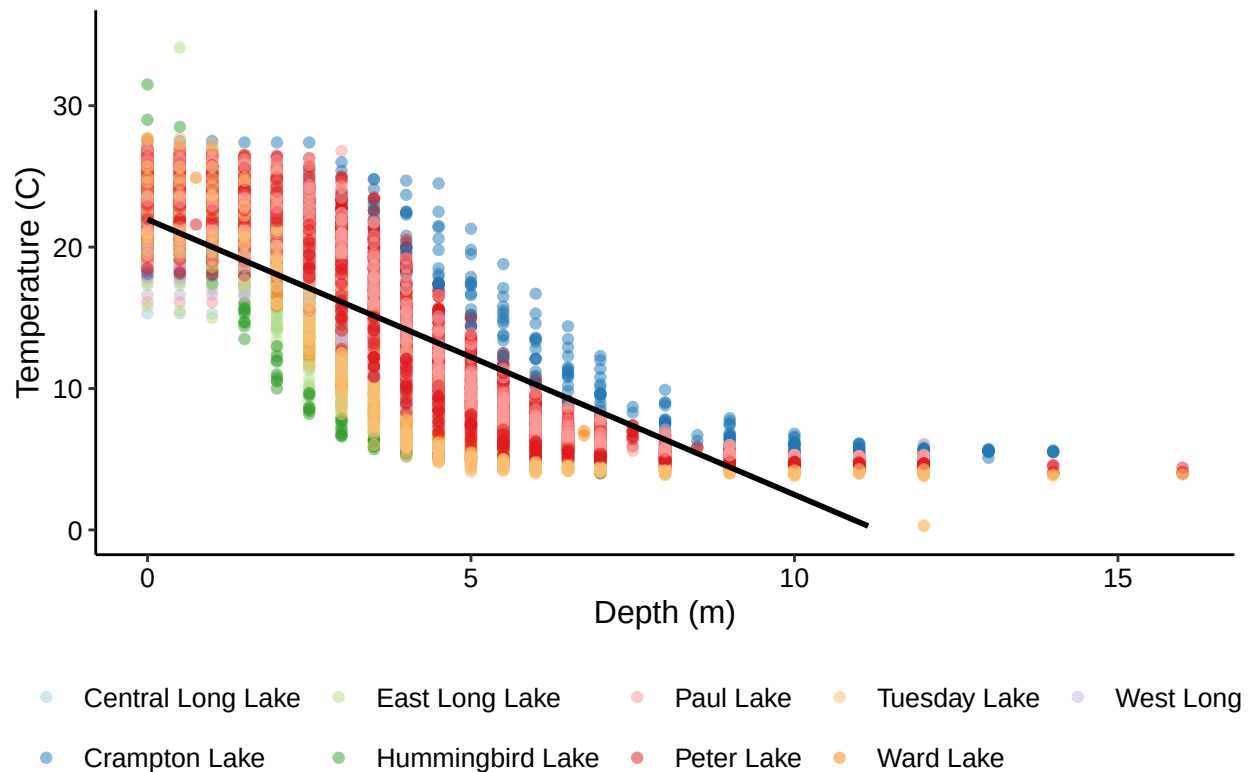
14. Create a graph that depicts temperature by depth, with a separate color for each lake. Add a `geom_smooth` (method = "lm", se = FALSE) for each lake. Make your points 50 % transparent. Adjust your y axis limits to go from 0 to 35 degrees. Clean up your graph to make it pretty.

```
#14.
ggplot(ntl_lter_subset, aes(x = depth, y = temperature_C)) +
  geom_point(alpha = 0.5, aes(color = lakename)) +
  geom_smooth(method = "lm", se = FALSE, color = "black") +
  scale_color_brewer(palette = "Paired") +
  scale_y_continuous(limits = c(0, 35)) +
  labs(x = "Depth (m)",
       y = "Temperature (C)",
       title = "Relationship Between Lake Depth and Temperature by Lake Name",
       color = NULL)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 24 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```

Relationship Between Lake Depth and Temperature by Lake Name



15. Use the Tukey's HSD test to determine which lakes have different means.

#15

```
TukeyHSD(lake_temp_anova)
```

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = temperature_C ~ lakename, data = ntl_lter_subset)
##
## $lakename
##
```

	diff	lwr	upr	p adj
Crampton Lake-Central Long Lake	-2.3145195	-4.7031913	0.0741524	0.0661566
East Long Lake-Central Long Lake	-7.3987410	-9.5449411	-5.2525408	0.0000000
Hummingbird Lake-Central Long Lake	-6.8931304	-9.8184178	-3.9678430	0.0000000
Paul Lake-Central Long Lake	-3.8521506	-5.9170942	-1.7872070	0.0000003
Peter Lake-Central Long Lake	-4.3501458	-6.4115874	-2.2887042	0.0000000
Tuesday Lake-Central Long Lake	-6.5971805	-8.6971605	-4.4972005	0.0000000
Ward Lake-Central Long Lake	-3.2077856	-6.1330730	-0.2824982	0.0193405
West Long Lake-Central Long Lake	-6.0877513	-8.2268550	-3.9486475	0.0000000
East Long Lake-Crampton Lake	-5.0842215	-6.5591700	-3.6092730	0.0000000
Hummingbird Lake-Crampton Lake	-4.5786109	-7.0538088	-2.1034131	0.0000004
Paul Lake-Crampton Lake	-1.5376312	-2.8916215	-0.1836408	0.0127491
Peter Lake-Crampton Lake	-2.0356263	-3.3842699	-0.6869828	0.0000999
Tuesday Lake-Crampton Lake	-4.2826611	-5.6895065	-2.8758157	0.0000000

```
## Ward Lake-Crampton Lake      -0.8932661 -3.3684639  1.5819317 0.9714459
## West Long Lake-Crampton Lake -3.7732318 -5.2378351 -2.3086285 0.0000000
## Hummingbird Lake-East Long Lake  0.5056106 -1.7364925  2.7477137 0.9988050
## Paul Lake-East Long Lake      3.5465903  2.6900206  4.4031601 0.0000000
## Peter Lake-East Long Lake      3.0485952  2.2005025  3.8966879 0.0000000
## Tuesday Lake-East Long Lake    0.8015604 -0.1363286  1.7394495 0.1657485
## Ward Lake-East Long Lake      4.1909554  1.9488523  6.4330585 0.0000002
## West Long Lake-East Long Lake  1.3109897  0.2885003  2.3334791 0.0022805
## Paul Lake-Hummingbird Lake     3.0409798  0.8765299  5.2054296 0.0004495
## Peter Lake-Hummingbird Lake    2.5429846  0.3818755  4.7040937 0.0080666
## Tuesday Lake-Hummingbird Lake  0.2959499 -1.9019508  2.4938505 0.9999752
## Ward Lake-Hummingbird Lake     3.6853448  0.6889874  6.6817022 0.0043297
## West Long Lake-Hummingbird Lake 0.8053791 -1.4299320  3.0406903 0.9717297
## Peter Lake-Paul Lake           -0.4979952 -1.1120620  0.1160717 0.2241586
## Tuesday Lake-Paul Lake         -2.7450299 -3.4781416 -2.0119182 0.0000000
## Ward Lake-Paul Lake            0.6443651 -1.5200848  2.8088149 0.9916978
## West Long Lake-Paul Lake       -2.2356007 -3.0742314 -1.3969699 0.0000000
## Tuesday Lake-Peter Lake        -2.2470347 -2.9702236 -1.5238458 0.0000000
## Ward Lake-Peter Lake           1.1423602 -1.0187489  3.3034693 0.7827037
## West Long Lake-Peter Lake      -1.7376055 -2.5675759 -0.9076350 0.0000000
## Ward Lake-Tuesday Lake         3.3893950  1.1914943  5.5872956 0.0000609
## West Long Lake-Tuesday Lake    0.5094292 -0.4121051  1.4309636 0.7374387
## West Long Lake-Ward Lake       -2.8799657 -5.1152769 -0.6446546 0.0021080
```

```
HSD_groups <- HSD.test(lake_temp_anova, "lakename", group = TRUE)
HSD_groups$groups
```

```
##           temperature_C groups
## Central Long Lake      17.66641      a
## Crampton Lake          15.35189     ab
## Ward Lake              14.45862     bc
## Paul Lake              13.81426      c
## Peter Lake             13.31626      c
## West Long Lake         11.57865      d
## Tuesday Lake           11.06923     de
## Hummingbird Lake       10.77328     de
## East Long Lake         10.26767      e
```

16. From the findings above, which lakes have the same mean temperature, statistically speaking, as Peter Lake? Does any lake have a mean temperature that is statistically distinct from all the other lakes?

Answer: According to the Tukey's HSD test, assuming an alpha of 0.05, Peter Lake has statistically the same mean temperature as Ward Lake and Paul Lake.

17. If we were just looking at Peter Lake and Paul Lake. What's another test we might explore to see whether they have distinct mean temperatures?

Answer: If we are only considering these two lakes, we would be able to use a two-sample t-test.

18. Wrangle the July data to include only records for Crampton Lake and Ward Lake. Run the two-sample T-test on these data to determine whether their July temperature are same or different. What does the test say? Are the mean temperatures for the lakes equal? Does that match your answer for part 16?

```

# filter to just Crampton Lake and Ward Lake
crampton_ward_lake <- ntl_lter_subset %>%
  filter(lakename %in% c("Crampton Lake", "Ward Lake"))

# run t-test
t.test(crampton_ward_lake$temperature_C ~ crampton_ward_lake$lakename)

##
## Welch Two Sample t-test
##
## data:  crampton_ward_lake$temperature_C by crampton_ward_lake$lakename
## t = 1.1181, df = 200.37, p-value = 0.2649
## alternative hypothesis: true difference in means between group Crampton Lake and group Ward Lake is not equal to 0
## 95 percent confidence interval:
##  -0.6821129  2.4686451
## sample estimates:
## mean in group Crampton Lake      mean in group Ward Lake
##                15.35189                14.45862

```

Answer: The t-test found a p-value of 0.2649, which does not allow us to reject our null hypothesis that the mean July temperature is the same between Crampton Lake and Ward Lake. The sample means are 15.35 degrees C and 14.46 degrees C for Crampton Lake and Ward Lake, respectively. Even though these means are not equal, the t-test indicates that true mean temperatures of these two lakes are statistically the same. This matches the answer for part 16: the Tukey test found a p-value of 0.971 for the pair and assigned both lakes to the same group, so it confirms that these two lakes have statistically the same mean.